Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/12/2025
PI Name:	Lau, Anthony G Ph.D.		
Project Title:	Effects of Acute and Protracted Galactic Cosmic Radiation on Bone Strength		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	LauA@tcnj.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	609-771-2644
Organization Name:	College of New Jersey		
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City:	Ewing	**State:**	NJ
Zip Code:	08618-1104	**Congressional District:**	12
Comments:	NOTE: As of Fall 2015, Dr. Lau is at The College of New Jersey. Previously at University of North Carolina at Chapel Hill while NSBRI postdoc.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	06/30/2021	**End Date:**	04/14/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/14/2025 per NSSC information (Ed., 10/12/24). NOTE: End date changed to 06/29/2024 per NSSC information (Ed., 7/20/23). NOTE: End date changed to 06/30/2023 per NSSC information (Ed., 8/5/22).		
Key Personnel Changes/Previous PI:	Alexis Mraz, Ph.D. was added to the research team per the PI; Dr. Mraz is at the College of New Jersey (Ed., 8/21/25).		
COI Name (Institution):	Davis, Catherine Ph.D. (Uniformed Services University of the Health Sciences)		
Grant/Contract No.:	80NSSC21K1506		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The exposure to ionizing space radiation can lead to damage of multiple biological tissue systems. The proposed work investigates degeneration of the central nervous system (CNS) and bone tissues from exposure to different does of simulated space radiation. Radiation is known to affect biological pathways that regulate both the CNS and bone. The objective of the proposed work to is to investigate the relationship between declines in cognitive function and declines from exposure to simulated space radiation, as well as to quantify these changes. The objectives will be addressed through animal studies exposing rats to acute and protracted (or fractionated) simulated galactic cosmic radiation and investigating the relationship between neurobehavioral deficits and bone degradation 7, 30, 90, and 180 days after radiation exposure. This tissue sharing proposal is part of an on-going collaboration between Dr. Catherine Davis at Uniformed Services University, who is currently funded by NASA to investigate the cognitive degradation in rats exposed to space radiation. Our lab has been collecting the hind limbs from her studies to investigate the corresponding bone strength changes in these rats. Neurobehavioral assessments include odor recognition memory tests and sustained attention tests. A multi-length scale approach will be performed to assess the corresponding bone health changes. Bone health assessments include microstructural (microCT scans), material property (micro-indentation), and whole bone (3-point bending) evaluations of bone strength. Analysis will be performed on CNS and bone endpoint measurements to determine whether the neurobehavioral deficits are predictive of declines in bone strength. The work is significant to NASA's goal for astronaut health during long duration spaceflight. Establishing a relationship between the CNS and bone response to radiation can provide valuable information for potential mechanisms and countermeasure targets for both systems. In the 2024-2025 performance period, we collected additional specimens for the GCR Study, completed all biomechanical testing and microCT scanning on these additional animals. And finished up microCT imaging of the skull caps from the high dose head-only radiation in a mini-pig animal model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides new insights on how lower doses of ionizing radiation (<0.5 Gy) affects bone health and strength. The skeletal changes from these lower doses could be considered for exposure to clinical radiation for diagnostic (i.e. CT Scans) and radiation therapies.
Task Progress:	During the past reporting period, additional animal specimens were added to the study from radiation exposure runs 24A/B at Brookhaven National Labs. Bone specimens were collected at 7, 30, and 180 day time points after exposures, which brings the total animals in this study to N=246. All experimental biomechanical testing (i.e., 3-point bending, microindentation) as well as microCT Scans for these specimens have been completed. The quantitative image process and computational modeling of the microCT image data are ongoing. The final statistical analysis of all the bone endpoint measurements is ongoing. The analysis to investigate the relationships between bone and the central nervous system response to radiation will be completed when the analysis of all the longitudinal neurobehavioral data from this final cohort of animals is completed. As part of Post-Doc Dr. Laura Bowman's NASA Human Research Program (HRP) Augmentation Award, blood ELISAs were performed to evaluate radiation effects on circulating biomarkers of bone metabolism. In addition, skeletal tissues from 12 minipigs exposed to head-directed radiation were analyzed with biomechanical testing and microCT imaging. We are currently waiting for the second half of the animal exposure study to be performed (12 additional minipigs) to complete the study.
Bibliography Type:	Description: (Last Updated: 08/20/2025)
Abstracts for Journals and Proceedings	Lau AG, Thomas PK, Connell RR, Mraz AL, Bowman LC, Davis CM. "Multi-organ animal model for investigating bone and CNS changes from low dose ionizing radiation exposure." 2024 Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 2024 Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Lau AG, Bowman LC, Perdikis P, Frolio S, Stibler AK, Cavallero NJ, Meyers MN, Vander Wiele SS, Felipe JT, Torres G, Connell RR, Thomas PK, Okulicz CJ, Johnson D, Mraz AL, Davis CM. "Investigating bone and CNS changes from low dose ionizing radiation exposure." 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Frolio S, Perdikis L, Stibler AK, Koch AL, Phan LM, Bagley BD, Rigor RR, Lau AG. "Mechanical and material properties of bone exposed to X-ray radiation." 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025